

RAFFLES INSTITUTION
2023 YEAR 6 JULY COMMON TEST

Time: 2 hr 30 min

H2 PHYSICS

9749

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Section A

INSTRUCTIONS TO CANDIDATES

There are 2 sections in this paper.

Section A consists of 15 multiple-choice questions. For each question, there are four possible answers **A**, **B**, **C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate Optical Mark Sheet.

You are advised to attempt Section A first. The Optical Mark Sheets will be collected after the first **30** minutes.

Section B consists of **6** structured questions and **1** data analysis question. You are to write your answers in the spaces provided.

Attempt ALL questions in Sections A and B.

Data

speed of light in free space
 permeability of free space
 permittivity of free space

 elementary charge
 the Planck constant
 unified atomic mass constant
 rest mass of electron
 rest mass of proton
 molar gas constant
 the Avogadro constant
 the Boltzmann constant
 gravitational constant
 acceleration of free fall

$$\begin{aligned}
 c &= 3.00 \times 10^8 \text{ m s}^{-1} \\
 \mu_0 &= 4\pi \times 10^{-7} \text{ H m}^{-1} \\
 \epsilon_0 &= 8.85 \times 10^{-12} \text{ F m}^{-1} \\
 &= (1/(36\pi)) \times 10^{-9} \text{ F m}^{-1} \\
 e &= 1.60 \times 10^{-19} \text{ C} \\
 h &= 6.63 \times 10^{-34} \text{ J s} \\
 u &= 1.66 \times 10^{-27} \text{ kg} \\
 m_e &= 9.11 \times 10^{-31} \text{ kg} \\
 m_p &= 1.67 \times 10^{-27} \text{ kg} \\
 R &= 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \\
 N_A &= 6.02 \times 10^{23} \text{ mol}^{-1} \\
 k &= 1.38 \times 10^{-23} \text{ J K}^{-1} \\
 G &= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} \\
 g &= 9.81 \text{ m s}^{-2}
 \end{aligned}$$

Formulae

uniformly accelerated motion

 work done on/by a gas
 hydrostatic pressure
 gravitational potential
 temperature

 pressure of an ideal gas

 mean translational kinetic energy of an ideal gas molecule

 displacement of particle in s.h.m.
 velocity of particle in s.h.m.
 electric current
 resistors in series
 resistors in parallel

 electric potential

 alternating current/voltage

 magnetic flux density due to a long straight wire

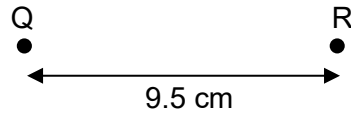
 magnetic flux density due to a flat circular coil

 magnetic flux density due to a long solenoid
 radioactive decay
 decay constant

$$\begin{aligned}
 s &= ut + \frac{1}{2}at^2 \\
 v^2 &= u^2 + 2as \\
 W &= p\Delta V \\
 p &= \rho gh \\
 \phi &= -Gm/r \\
 T/K &= T / ^\circ\text{C} + 273.15 \\
 p &= \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \\
 E &= \frac{3}{2} kT \\
 x &= x_0 \sin \omega t \\
 v &= v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2} \\
 I &= Anvq \\
 R &= R_1 + R_2 + \dots \\
 1/R &= 1/R_1 + 1/R_2 + \dots \\
 V &= \frac{Q}{4\pi\epsilon_0 r} \\
 x &= x_0 \sin \omega t \\
 B &= \frac{\mu_0 I}{2\pi d} \\
 B &= \frac{\mu_0 NI}{2r} \\
 B &= \mu_0 nI \\
 x &= x_0 \exp(-\lambda t) \\
 \lambda &= \ln 2 / t_{1/2}
 \end{aligned}$$

Section A

- 1 Two charged particles, Q and R, are fixed in space and separated by a distance of 9.5 cm.



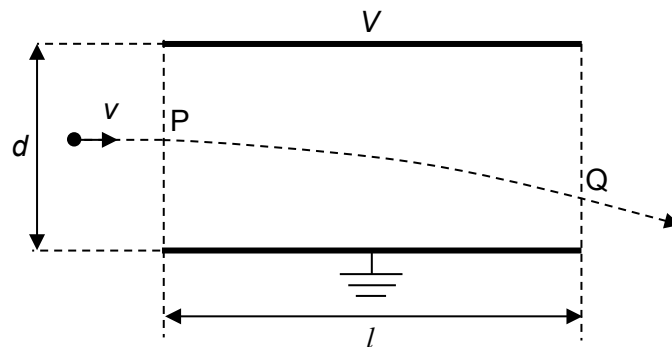
Particle Q has a charge of 8.6 nC and particle R has a charge of -4.5 nC.

What is the magnitude of the electric field strength midway between the charges?

- A** $2.5 \times 10^3 \text{ N C}^{-1}$ **B** $1.3 \times 10^4 \text{ N C}^{-1}$ **C** $1.6 \times 10^4 \text{ N C}^{-1}$ **D** $5.2 \times 10^4 \text{ N C}^{-1}$

- 2 A uniform electric field is set up in the region between two parallel plates of length l , separated by a distance d . The potential difference between the two plates is V .

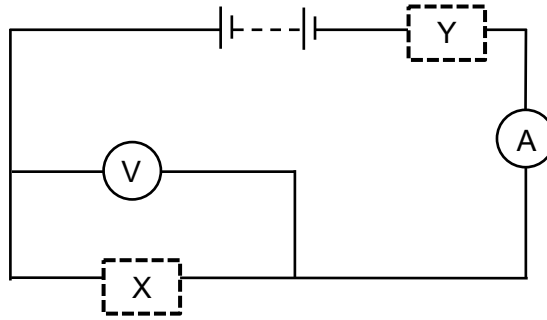
A particle of mass m , charge q and speed v enters the uniform field at right angles to the field. The particle enters and exits the field at points P and Q respectively.



What is the potential difference between P and Q?

- A** $\frac{qV^2l^2}{2md^2v^2}$ **B** $\frac{2md^2v^2}{ql^2}$ **C** $\frac{qV^2v^2}{2md^2l^2}$ **D** $\frac{ml^2}{2qv^2}$

- 3 Two electrical components X and Y are connected in a circuit as shown below. The readings on the voltmeter and ammeter are noted.

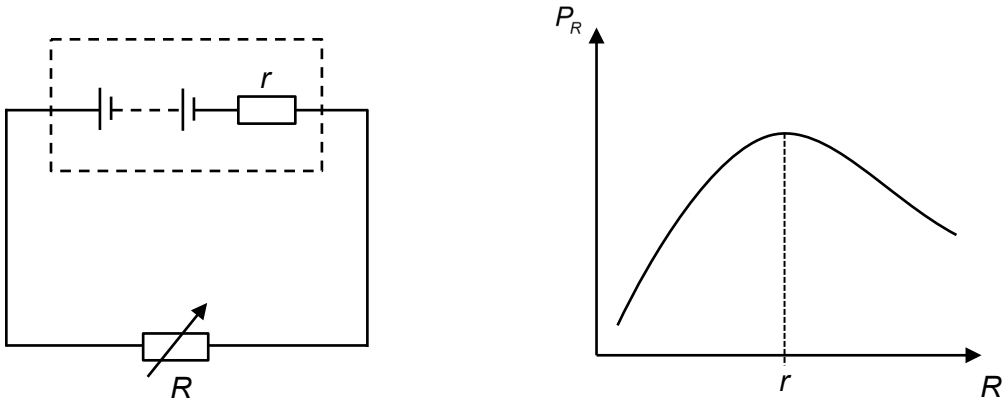


The temperature of the surroundings is then increased while the light intensity is decreased. It is observed that the reading on the voltmeter increases while the reading on the ammeter remains the same.

Which of the following is a possible combination of components X and Y?

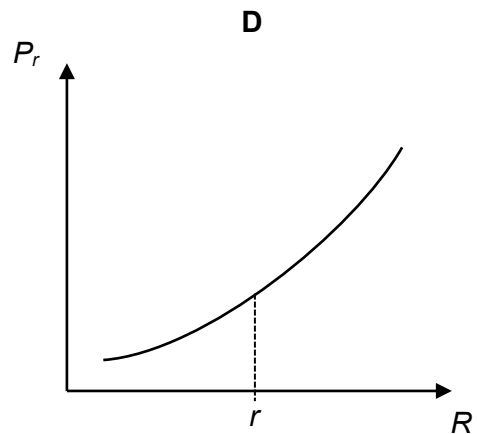
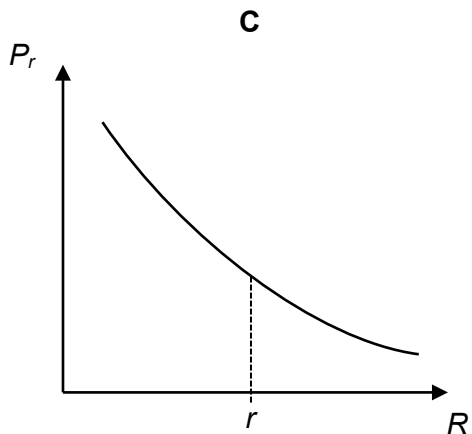
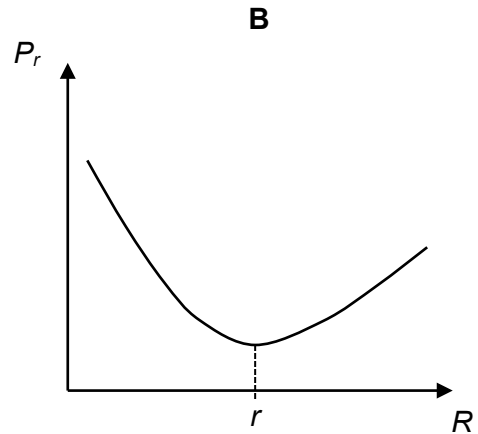
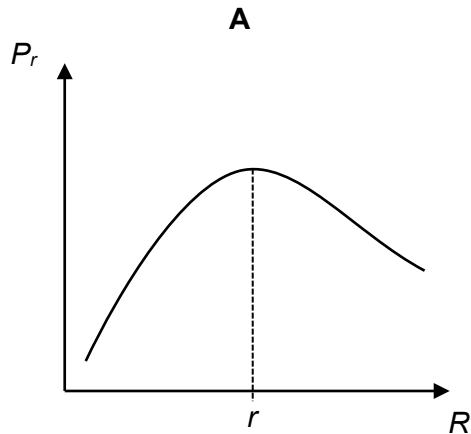
	X	Y
A	thermistor	light-dependent resistor
B	light-dependent resistor	thermistor
C	ohmic conductor	thermistor
D	light-dependent resistor	ohmic conductor

- 4 A battery with internal resistance r is connected to a variable resistor R as shown below.

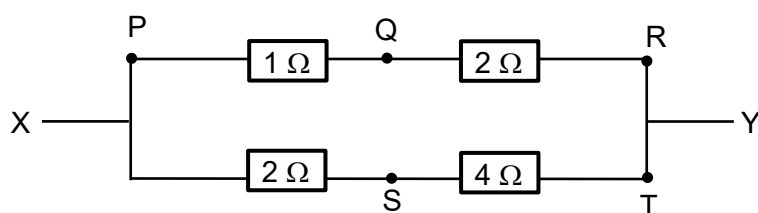


The graph shows the variation with R of the power P_R dissipated by R .

Which of the following graphs shows the variation with R of the power P_r dissipated by the internal resistance r ?



- 5 Four resistors are connected across points X and Y as shown.

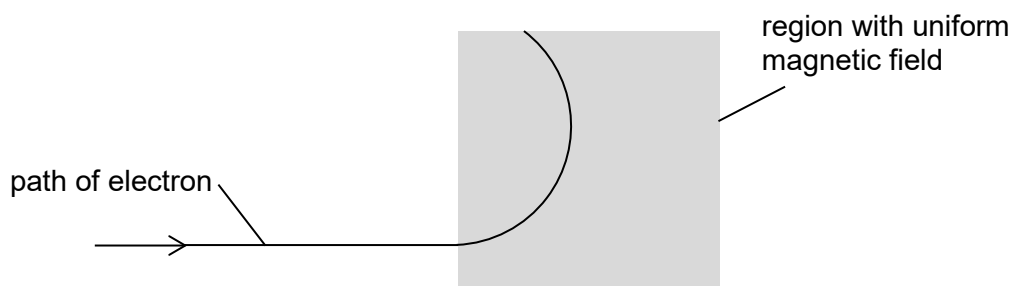


Another resistor of $3\ \Omega$ is connected across two points in the circuit.

Which of the following two points should it be connected across such that the effective resistance across XY is the smallest?

- A** PQ **B** PT **C** QR **D** QS
- 6 An electron, travelling in a straight line at $1.46 \times 10^7\ \text{m s}^{-1}$, enters a region where there is a uniform magnetic field.

The diagram shows the path of the electron before it enters the magnetic field as well as within the field.

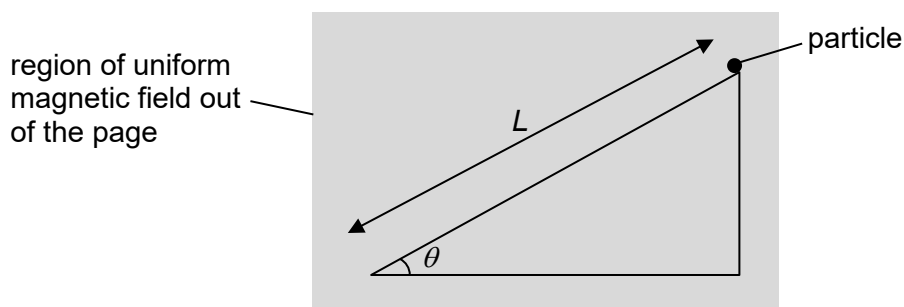


Within the magnetic field, the electron moves in a circular path of diameter $0.0700\ \text{m}$.

What is the direction of the magnetic field and the magnitude of the magnetic flux density?

	direction of magnetic field	magnitude of magnetic flux density / T
A	into page	1.19×10^{-3}
B	into page	2.38×10^{-3}
C	out of page	1.19×10^{-3}
D	out of page	2.38×10^{-3}

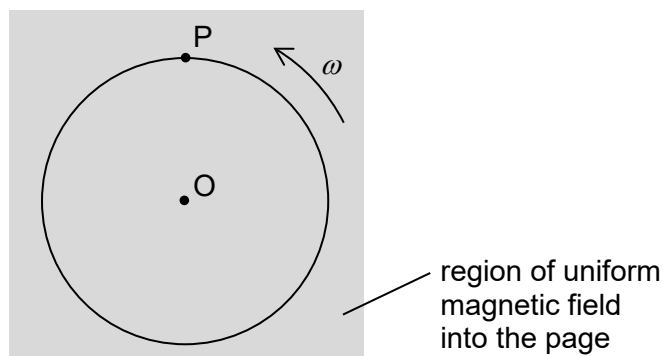
- 7 A particle of mass m and positive charge q is released from rest from the top of a smooth inclined plane of length L and angle of inclination θ as shown and slides down the plane.



The entire inclined plane is in a region of uniform magnetic field of flux density B directed out of the page.

What is the magnitude of the normal contact force exerted by the inclined plane on the particle when it is at the midpoint of the plane?

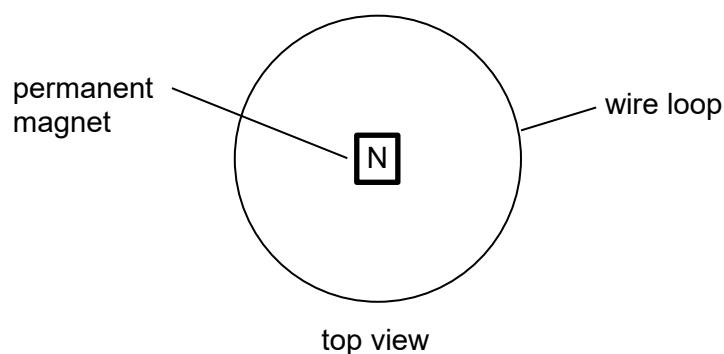
- A** $mg \cos \theta + Bq\sqrt{gL \sin \theta}$ **B** $mg \cos \theta - Bq\sqrt{gL \sin \theta}$
C $mg \sin \theta - Bq\sqrt{gL \sin \theta}$ **D** $mg \sin \theta - Bq\sqrt{gL \cos \theta}$
- 8 A conducting disc of radius r rotates about its centre in an anticlockwise direction at a constant angular velocity ω . It is placed in a region of uniform magnetic field perpendicular to its surface. A steady e.m.f. is generated between the centre O and the rim at P .



Which point has higher potential and what is the direction of the induced e.m.f.?

	higher potential	direction of induced e.m.f.
A	O	O to P
B	O	P to O
C	P	O to P
D	P	P to O

- 9 A permanent bar magnet with North pole at the top is dropped vertically into the page through a loop of wire.



Which of the following best describes the directions of the current induced in the loop when viewed from the top as the magnet enters and leaves the loop?

	when entering the loop	when leaving the loop
A	clockwise	clockwise
B	anticlockwise	anticlockwise
C	clockwise	anticlockwise
D	anticlockwise	clockwise

- 10 When a heating coil of resistance R carries a constant current of 2.4 A, a power P is generated.

Another heating coil of resistance $4R$ carries an alternating sinusoidal current and a mean power P is also generated.

What is the peak current in the second heating coil?

- A** 0.83 A **B** 0.85 A **C** 1.2 A **D** 1.7 A

- 11 In an ideal transformer, the r.m.s. potential difference across the primary coil is 12 V. The secondary coil is connected in series to a resistor of resistance $15\ \Omega$.

What is the r.m.s. current in the primary coil when the r.m.s. current in the secondary coil is 1.3 A?

- A** 0.47 A **B** 1.4 A **C** 1.6 A **D** 2.1 A

- 12** A metal surface is illuminated with a beam of monochromatic light.

What affects the maximum speed of the photoelectrons that are emitted from the surface?

- A** the frequency of the light and the work function of the metal
- B** the intensity and the frequency of the light
- C** the frequency of the light only
- D** the intensity of the light only

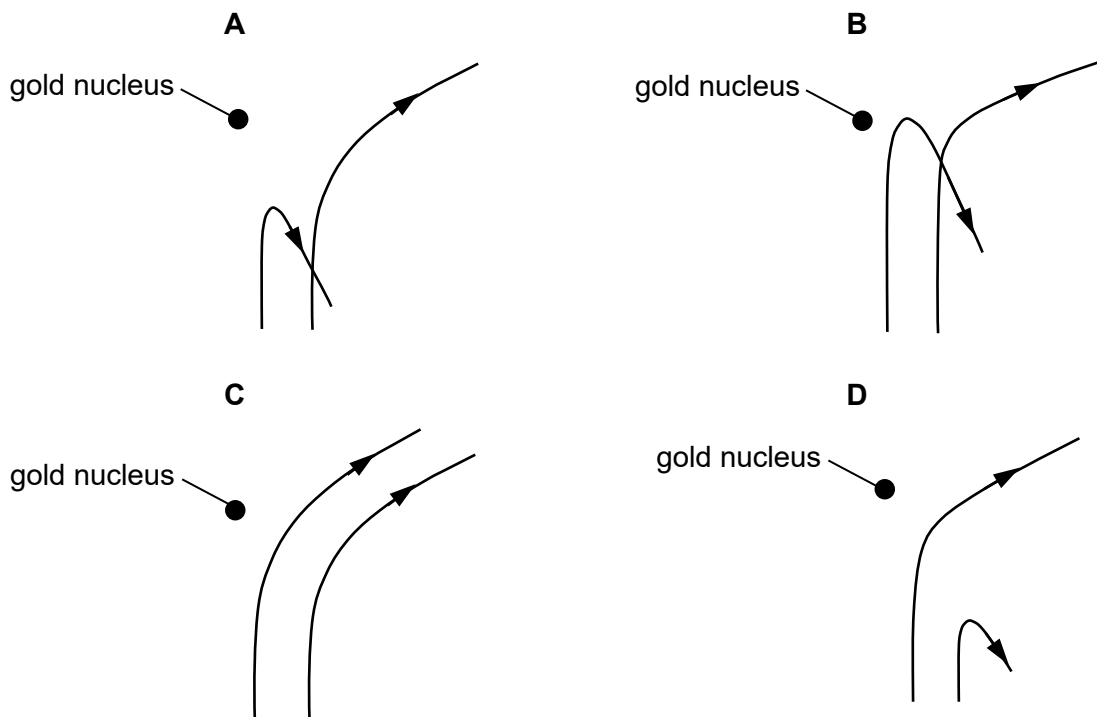
- 13** Electromagnetic radiation is detected when electrons accelerated through a potential difference of 30 kV collide with a metal target.

What is the shortest wavelength present in the radiation produced?

- A** 6.6×10^{-30} m **B** 2.8×10^{-21} m **C** 7.1×10^{-12} m **D** 4.1×10^{-11} m

- 14** In the Rutherford alpha particle scattering experiment, alpha particles are directed at a thin gold foil.

Which diagram best represents the paths of alpha particles scattered by a gold nucleus?



15 A detector of ionising radiation gives a background count rate of 17 counts per minute.

A radioactive source is placed close to the detector and the reading is 673 counts per minute.

What will be the reading after two half-lives of the source?

- A** 164 counts per min
- B** 168 counts per min
- C** 181 counts per min
- D** 185 counts per min